

Epinephrine's Multiple Effects: A Closer Look How can epinephrine coordinate a response to stress that involves widely varying effects in individual tissues? We can answer that question by examining different response pathways (**Figure 45.20**) in a range of target cells:

- In liver cells, epinephrine binds to a β -type receptor in the plasma membrane. This receptor activates the enzyme protein kinase A, which in turn regulates enzymes of glycogen metabolism, causing release of glucose into the blood (see **Figure 45.20a**). Note that this is the signal transduction pathway illustrated in **Figure 45.6**.
- In the smooth muscle cells that line blood vessels supplying skeletal muscle, the same kinase activated by the same epinephrine receptor inactivates a muscle-specific enzyme. The result is smooth muscle relaxation, leading to vasodilation and hence increased blood flow to skeletal muscles (see **Figure 45.20b**).
- In the smooth muscle cells lining blood vessels of the intestines, epinephrine binds to an α -type receptor (see **Figure 45.20c**). This receptor triggers a signaling pathway that involves enzymes other than protein kinase A and that causes smooth muscle contraction rather than

relaxation. The resulting vasoconstriction reduces blood flow to the intestines, facilitating the redirection of blood to active skeletal muscle.

Thus, epinephrine elicits multiple responses if its target cells differ in the receptor protein they express or in the molecules activated by the receptor upon hormone binding. As illustrated in these examples, such variation in response plays a key role in enabling epinephrine to trigger a range of activities that together bring about a coordinated rapid response to stressful stimuli.

The Role of the Adrenal Cortex

Like the adrenal medulla, the adrenal cortex mediates an endocrine response to stress (see **Figure 45.19b**). The two portions of the adrenal gland differ, however, in both the types of stress that trigger a response and the targets of the hormones that are released.

The adrenal cortex becomes active under stressful conditions that include low blood sugar, decreased blood volume and pressure, and shock. Such stimuli cause the hypothalamus to secrete a releasing hormone that stimulates the anterior pituitary to release adrenocorticotropic hormone

(ACTH), a tropic hormone. When ACTH reaches the adrenal cortex via the bloodstream, it stimulates the endocrine cells to synthesize and secrete a family of steroids called *corticosteroids*. The two main types of corticosteroids in humans are glucocorticoids and mineralocorticoids.

Glucocorticoids, such as cortisol (see **Figure 45.4**), make more glucose available as fuel by promoting glucose synthesis from noncarbohydrate sources, such as proteins. Glucocorticoids also act on skeletal muscle, causing the breakdown of muscle proteins into amino acids. These are transported to the liver and kidneys, converted to glucose, and released into the blood. The synthesis of glucose upon the breakdown of muscle proteins provides circulating fuel when the body requires more glucose than the liver can mobilize from its glycogen stores.

If glucocorticoids are introduced into the body at a level above that normally present, they suppress certain components of the body's immune system. For this reason, glucocorticoids are sometimes used to treat inflammatory diseases such as arthritis. However, their long-term use can have serious side effects on metabolism. Nonsteroidal anti-inflammatory drugs (NSAIDs), such as

▼ **Figure 45.20 One hormone, different effects.** Epinephrine, the primary “fight-or-flight” hormone, produces different responses in different target cells. Target cells with the same receptor exhibit different responses if they have different signal transduction pathways or effector proteins; compare **(a)** with **(b)**. Target cells with different receptors for the hormone often exhibit different responses; compare **(b)** with **(c)**.

